

**Subject : DS-505: Big Data Handling and Management for Machine Learning Applications**  
**Course : Bachelors of Science in Data Science and Analytics Semester : Semester-5**  
**Model Paper 1**

|       |                                                                                                                                                                                                                 |          |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Q.1 A | Answer the following in short (Any 4)                                                                                                                                                                           | 04 Marks |
| 1     | Adding more RAM, bigger SSDs, or faster CPUs to an existing single computer is an example of:<br>a) Horizontal Scaling (Scale-Out) b) Vertical Scaling (Scale-Up)<br>c) Distributed Computing d) Grid Computing |          |
| 2     | In Apache Spark, which of the following operations is an Action that triggers DAG execution?<br>a) df.select() b) df.filter()<br>c) df.count() d) df.groupBy()                                                  |          |
| 3     | Which technique converts categorical text labels into unique integer numbers (0, 1, 2, ...)?<br>a) StandardScaler b) LabelEncoder<br>c) SimpleImputer d) TrainTestSplit                                         |          |
| 4     | The process of breaking down a text sentence into individual words or subwords in NLP is called:<br>a) Lemmatization b) Tokenization<br>c) Vectorization d) Stemming                                            |          |
| 5     | Which file format stores data in a binary columnar format with built-in compression?<br>a) CSV b) JSON<br>c) Apache Parquet d) Plain Text                                                                       |          |
| Q.1 B | Answer the following in short (Any 4)                                                                                                                                                                           | 04 Marks |
| 1     | What is the primary difference between Structured and Unstructured data?                                                                                                                                        |          |
| 2     | What is the role of the Driver Program in Apache Spark?                                                                                                                                                         |          |
| 3     | Define Data Leakage in machine learning preprocessing.                                                                                                                                                          |          |
| 4     | Mention any two Python string functions used for basic text cleaning.                                                                                                                                           |          |
| 5     | What is the default port number used to access the Apache Spark Web UI?                                                                                                                                         |          |
| Q.2 A | Write a short note on the following (any one)                                                                                                                                                                   | 05 Marks |
| 1     | Differentiate between Traditional Relational Databases (RDBMS) and Big Data Systems with suitable comparison points (such as scaling, schema, and data types).                                                  |          |
| 2     | Describe the Apache Spark Distributed Architecture with a neat labeled diagram, detailing the functions of the Driver Node, Cluster Manager, and Worker Executors.                                              |          |
| B     | Explain any four basic PySpark DataFrame operations from the following with their purpose and simple syntax:<br>i) df.show() ii) df.select() iii) df.filter() iv) df.count() v) df.groupBy()                    | 04 Marks |
| Q.3   | Write a short note on the following (any Two)                                                                                                                                                                   | 08 Marks |
| 1     | Explain the common techniques used for handling Missing Values (dropping rows/columns, mean, median, and mode imputation) and Duplicate Records in a dataset with suitable examples.                            |          |
| 2     | What is a Chatbot? Explain how Large Language Models (LLMs) help in building intelligent customer support chatbots.                                                                                             |          |
| 3     | Explain the working principle of the K-Nearest Neighbors (KNN) classifier with a simple diagram. Why is it called a "Lazy Learner"?                                                                             |          |

